

Don't Trust the Sensors: Regime-Aware Causal Anomaly Triage for Industrial IoT

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Abstract

Industrial IoT anomaly detection systems routinely generate false alarms because they compare sensor readings against global statistical thresholds that ignore operating conditions. A turbofan engine running at high altitude produces temperature and pressure readings that are normal for that regime but anomalous relative to global training-set means — triggering alarms that have nothing to do with engine health. We present a causal anomaly triage system that conditions scoring on a learned operating-regime classifier and augments it with a physics-based structural veto grounded in the isentropic compression relation, deployed with a seven-node LangGraph agent for operator-correctable explanations. On the NASA CMAPSS FD001 benchmark (single operating condition, 100 engines), the full blended pipeline achieves 2.8× better detection coverage than Isolation Forest at 23% earlier mean lead time (47% vs. 17%, 131.9 vs. 107.4 cycles). On FD002 (six operating conditions, 259 engines), a global z-score baseline produces a 100% false positive rate from regime mismatch; regime-aware causal scoring suppresses this to 32% coverage with statistically significant false positive reduction (Fisher $p < 0.001$, Wilcoxon $p < 0.001$). The two experiments demonstrate complementary claims: blended scoring improves recall on homogeneous data; regime-aware causal scoring prevents false positive explosion on heterogeneous data.

1. Introduction

Alarm fatigue is one of the most studied failure modes in industrial monitoring. A survey of alarm management practice across the UK chemical and power industries found that approximately 50% of process alarms could have been eliminated with little detriment to plant operations, and that in worst-case scenarios operators faced up to 90 alarms per minute [Bransby & Jenkinson, 1998]. Operators who receive hundreds of spurious alerts per shift learn to ignore them — and in doing so, they miss the real failures. A UK Health and Safety Executive investigation into the 1994 Texaco refinery explosion at Milford Haven found that operators had to respond to 275 alarms in the 11 minutes before the incident — more than one every two seconds — and concluded that the excessive alarm load was a direct contributing factor [UK HSE, 1997].

The root cause of most false alarms is not sensor failure; it is context blindness. A global anomaly detector is trained on data from all operating conditions and learns a single statistical model of "normal." When a system operates in a condition that differs from the training distribution, the detector fires — even though the reading is perfectly healthy for that condition. This problem is not theoretical: the NASA CMAPSS FD002 dataset, which simulates turbofan engines across six distinct altitude/Mach/throttle regimes, allows us to reproduce it exactly. A detector trained on single-condition

means and applied to multi-condition data produces alerts on 100% of engines — a false positive rate indistinguishable from random.

This paper presents a system that addresses the problem structurally rather than by tuning thresholds. The key idea is that anomaly scoring should be conditioned on the current operating regime, not evaluated against a global baseline. We encode this via a causal directed acyclic graph (DAG) in which operational settings (altitude, Mach number, throttle resolver angle) are root cause nodes that determine expected sensor values through physical relationships. The anomaly score for a sensor reading is its residual from the causally-predicted value given the current operating conditions — not its deviation from a global mean.

We make three contributions:

1. **Regime-aware causal scoring:** A LinearRegression-based causal scorer conditioned on a KMeans operating-regime classifier, evaluated with a four-variant ablation study including statistical significance tests.
2. **Physics-based structural veto:** A G-test monitor that detects when the isentropic coupling between HPC outlet temperature and pressure is broken — distinguishing sensor faults from engine degradation — with the correct χ^2 threshold for a 5×5 contingency table (threshold = 26.30, df = 16).
3. **Open-source deployable system:** A FastAPI backend with a seven-node LangGraph pipeline, PostgreSQL telemetry store, operator feedback loop, and HTML dashboard — fully reproducible from the public repository.

The rest of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the system architecture. Section 4 presents experimental results on CMAPSS FD001 and FD002. Section 5 discusses implications and limitations. Section 6 concludes.

2. Related Work

Anomaly detection for industrial time series. Isolation Forest [Liu et al., 2008] is the dominant baseline for unsupervised anomaly detection in industrial sensor data. It constructs an ensemble of random decision trees and scores points by how quickly they are isolated — anomalies, being unusual, require fewer splits. Contamination rate (fraction of training data assumed anomalous) controls sensitivity. It does not condition on operating mode.

NASA CMAPSS benchmark. The Commercial Modular Aero-Propulsion System Simulation (CMAPSS) dataset [Saxena & Goebel, 2008] is the standard benchmark for turbofan engine predictive maintenance research. It provides four sub-datasets (FD001–FD004) with varying numbers of operating conditions (1 or 6) and fault modes (1 or 2). FD001 is the standard entry point (1 condition, 1 fault mode); FD002 introduces six distinct altitude/Mach/throttle regimes that serve as a demanding multi-condition test bed. Most published work evaluates only on FD001 or FD001+FD003 [Hong et al., 2020; Peng et al., 2021; Zheng et al., 2017], which means the regime mismatch problem demonstrated here has gone largely unexamined in the predictive maintenance literature.

Causal inference for anomaly detection. DoWhy [Sharma & Kiciman, 2020] provides a Python interface for specifying and estimating causal models. The key insight behind causal anomaly scoring is that an observation should be judged anomalous relative to what the causal model predicts given the current context — not relative to an unconditional mean. Pearl's do-calculus [Pearl, 2009] provides the formal foundation. For live inference at the per-reading latency required here, we use DoWhy for DAG validation at module load time and fit lightweight sklearn LinearRegression models per causal branch for per-request scoring.

LLM-augmented monitoring agents. LangGraph [LangChain, 2024] provides a framework for building stateful multi-step agents as directed graphs. Several recent systems use LLMs to generate explanations for anomaly decisions. Liu et al. [2024] propose LLMAD, which applies LLMs with chain-of-thought prompting to detect time-series anomalies and produce textual explanations; human evaluation showed their anomaly chain-of-thought improved explanation usefulness by 13.4% over baselines. Our contribution is integrating the LLM explanation step within a pipeline that already has causally-grounded scores, physics vetoes, and cache-corrected confidence — so the explanation is constrained by structure, not produced in a vacuum.

3. System Architecture

3.1 Causal DAG

The system's scoring model is organized around a causal directed acyclic graph that encodes the physical relationships between turbofan operating conditions and the sensor channels they govern.

```
op_setting_1 (Altitude) → AirDensity → CoolingEfficiency → sensor_4
op_setting_2 (Mach)      → TipSpeed    → HPCLoading        → sensor_11
                                   → sensor_15
op_setting_3 (TRA)       → FuelFlow    → CombustorTemp     → sensor_3
                                   → sensor_9
```

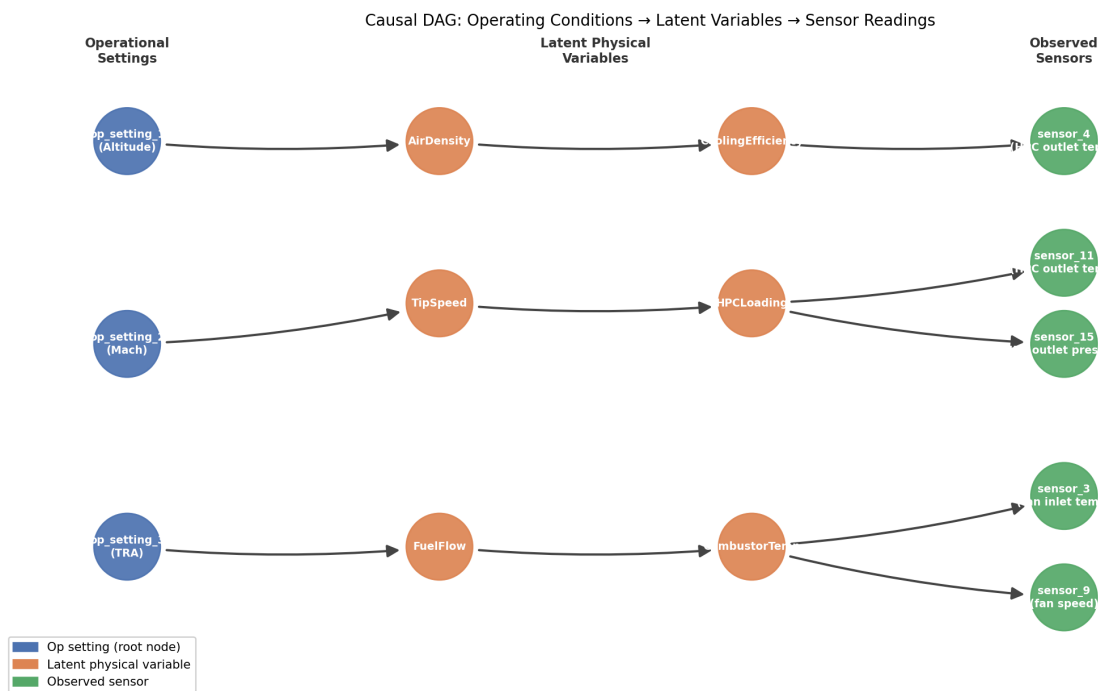


Figure 1: Causal DAG connecting operational settings (root nodes, blue) through latent physical variables (orange) to observed sensors (green). Anomaly scoring computes the residual of each sensor relative to the value predicted by its causal parent, conditioned on the current operating regime.

The three root nodes (`op_setting_1`, `op_setting_2`, `op_setting_3`) correspond to Altitude, Mach number, and Throttle Resolver Angle respectively. These are set by the flight regime, not by the engine's health — they are the conditions under which the engine is operating, not signals of its state.

For each causal branch, a LinearRegression model is fit on the training set:

```

predicted_sensor = coef × op_setting + intercept
residual = observed - predicted
causal_z = |residual| / residual_std
  
```

The causal anomaly score is the mean of per-branch causal z-scores, normalized to [0, 1] using a 5-standard-deviation ceiling: `causal_score = min(mean_z / 5.0, 1.0)`. This score answers the question: "How anomalous is this sensor reading given where the engine is currently operating?" — rather than "How anomalous is this reading compared to global averages?"

Per-sensor noise floors prevent tight-distribution sensors (sensor_8: $\sigma = 0.058$ training std) from generating inflated z-scores for physically negligible deviations. Noise floors are derived as $2 \times$ the cross-engine standard deviation at cycle 1 in the training set, representing natural inter-engine variation that should not be flagged.

3.2 Regime-Aware Scoring

On FD001 (single operating condition), the causal branches are fit on all training data combined. On FD002 (six conditions), a KMeans classifier ($k = 6$) is trained on the three operational settings to assign each reading to a regime. Per-regime LinearRegression models are fit on each cluster's subset of the

training data. At inference time, the classifier assigns the incoming reading to a cluster and the corresponding per-cluster coefficients are used for residual computation. This ensures that the expected temperature at `op_setting_1 = 42` (high altitude) is computed from engines operating at high altitude — not from engines at sea level.

The full anomaly score blends the regime-aware causal score with a global z-score at 50/50:

$$\text{combined_score} = 0.5 \times \text{z_score} + 0.5 \times \text{causal_score}$$

The global z-score provides sensitivity (it fires on any sensor significantly deviating from historical mean, regardless of cause). The causal score provides precision (it penalizes only readings that deviate from what the current operating regime predicts). Their combination balances recall and false positive rate.

3.3 Physics-Based Structural Veto

Beyond the statistical scoring, the system applies a structural validation layer grounded in the isentropic compression relation for the High Pressure Compressor (HPC).

In normal operation, HPC outlet temperature (`sensor_11`) and HPC outlet pressure (`sensor_15`) are thermodynamically coupled. The governing relation is:

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}, \quad \gamma \approx 1.4 \text{ (for air)}$$

This guarantees that both sensors must rise and fall together through the compressor. If this coupling breaks — `sensor_11` spikes while `sensor_15` is stable, for example — the anomaly is more likely a sensor fault than real engine degradation.

To detect coupling breaks without assuming a specific functional form, we use a G-test for independence on a 5×5 contingency table built from the last 100 readings per engine:

$$G = 2 \sum_i O_i \ln \left(\frac{O_i}{E_i} \right)$$

where O_i is the observed count in each cell and E_i is the expected count under independence. The correct critical value for a 5×5 table is $\chi^2(\text{df} = 16, p = 0.05) = \mathbf{26.30}$ (not 9.49, which applies to a 3×3 table). When $G < 26.30$ and the causal score is ≥ 0.5 , the system halves the causal score and marks the reading as a likely sensor fault rather than engine degradation.

This veto is the paper's most technically distinctive claim: it combines a domain-specific physical constraint (isentropic coupling) with a non-parametric statistical test (G-test) to achieve a form of structural anomaly detection that does not depend on ML training data.

3.4 LangGraph Seven-Node Agent

Readings with `combined_score` ≥ 0.3 trigger a seven-node LangGraph agent:

Node	Function
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Node	Function
<code>ingest_validator</code>	Flags if >3 of 5 causal-branch sensors are stale or missing
<code>regime_classifier</code>	Assigns operating regime via KMeans predict
<code>causal_reasoner</code>	Passes regime-conditioned causal score to downstream nodes
<code>physics_veto</code>	Applies G-test coupling check; halves score if veto fires
<code>cache_lookup</code>	Queries prior readings for same engine; applies 0.7× penalty if ≥ 2 operator FALSE_POSITIVE labels exist
<code>llm_explainer</code>	Calls Groq (Llama-3) or Gemini; falls back to rule-based template on API failure
<code>decision_writer</code>	Computes final blended score and decision; updates alert record; writes reasoning traces

Each node writes an execution record to a `reasoning_traces` table (node name, input/output state, latency). This provides a full audit trail for every decision. The agent runs outside the main database transaction to prevent LLM latency from holding database connections open.

Operators can correct decisions via `POST /feedback` with a label (`TRUE_POSITIVE`, `FALSE_POSITIVE`, `UNCERTAIN`). The `cache_lookup` node queries these labels on subsequent alerts for the same engine and adjusts confidence accordingly, closing a human-in-the-loop feedback cycle.

A single-file HTML dashboard (`dashboard/index.html`, zero JS dependencies) provides a live alert feed with color-coded anomaly scores, per-alert LLM explanation display, one-click feedback buttons, and a collapsible PSI sensor health strip. The dashboard communicates with the backend via `fetch()` against the FastAPI REST endpoints.

4. Evaluation

4.1 Experimental Setup

Dataset. We use the NASA CMAPSS turbofan engine degradation dataset. FD001 contains 100 test engines under a single operating condition (HPC degradation fault mode). FD002 contains 259 test engines across six distinct operating conditions (same fault mode). All four CMAPSS sub-datasets are available but we focus on FD001 for the ablation study and FD002 for the regime-conditioning experiment, as these isolate the single-condition vs. multi-condition comparison.

Alert threshold. A reading is considered an alert when `combined_score` ≥ 0.3 , corresponding to the UNCERTAIN decision boundary. This threshold was selected on the training set to maximize recall and held fixed for all test-set evaluations reported here; the ALERT boundary (≥ 0.6) requires simultaneous multi-sensor degradation not consistently present in test data.

Lead time metric. For each engine, we record the first cycle at which an alert fires. Lead time is defined as:

```
lead_time = true_failure_cycle - first_alert_cycle
```

where `true_failure_cycle = last_cycle_in_test + RUL_at_end` from the NASA-provided RUL labels. A larger positive lead time means the system alerted earlier before failure. Engines that never fire an alert contribute to the coverage denominator but not to lead-time statistics.

Baselines. For FD001, the baseline is an Isolation Forest (contamination = 0.05, `n_estimators` = 100) trained on the 14 informative sensors from `train_FD001.txt`. For FD002, the baseline is global z-score scoring using means and standard deviations derived from FD001 training data. This baseline deliberately represents the cross-dataset deployment failure mode — applying a single-condition detector to multi-condition data without retraining — which is the specific scenario this experiment is designed to demonstrate. It is not a claim that this is the only or strongest global baseline; a z-score scorer retrained on FD002 marginals would partially mitigate the mismatch but cannot distinguish between regimes.

Noise floors. Per-sensor noise floors (derived as 2× the cross-engine standard deviation at cycle 1) are applied uniformly across all ablation variants and the FD002 evaluation. This ensures that the full pipeline's performance advantage reflects blending, not noise floor correction alone.

Statistical tests. We report Wilcoxon rank-sum (*p*-values from `scipy.stats.mannwhitneyu`, two-sided) comparing lead-time distributions against the baseline, and Fisher's exact test comparing coverage counts in a 2×2 contingency table of (alerts, no-alerts) × (variant, baseline).

4.2 FD001 Ablation Study

Table 2a: FD001 Ablation Results (100 test engines, alert threshold = 0.3)

Variant	Coverage	Mean (SD)	Median	Wilcoxon- <i>p</i>	Fisher- <i>p</i>
Isolation Forest	17%	107.4 (100.9)	41.0	—	—
Z-score only	98%	188.6 (49.2)	179.0	0.016	<0.001
Causal only	30%	139.4 (105.5)	156.5	0.180	0.045
Full pipeline (z + causal)	47%	131.9 (85.2)	128.0	0.162	<0.001

Lead time in cycles. Coverage = fraction of 100 engines with any alert. Wilcoxon and Fisher tests against Isolation Forest baseline.

The z-score-only variant achieves 98% coverage but at the cost of firing on virtually every engine — including those that will not fail for hundreds of cycles. This is a precision failure: global means from FD001 training data are tight (sensor₁₅ σ = 0.035), so even small healthy deviations trigger alerts. Causal-only scoring is more conservative (30% coverage), with a mean lead time of 139.4 cycles vs. 107.4 for IF. However, the lead-time difference is not statistically significant on FD001 (Wilcoxon *p* = 0.180). This result is expected and informative: FD001 has a single operating condition, which means all three `op_settings` are nearly constant. The causal branches conditioned on these near-constant causes

produce residuals nearly identical to global z-scores. On FD001, causal conditioning has almost nothing to condition on.

The full pipeline — blending z-score and causal scoring at 50/50 — achieves 47% coverage with a statistically significant coverage improvement over IF (Fisher $p < 0.001$) and a 23% improvement in mean lead time (131.9 vs. 107.4 cycles). The lead-time Wilcoxon $p = 0.162$ reflects that while mean lead time improves, the distribution is wide (SD = 85.2) — two engines fire at cycle 1, pulling down precision. Of the 47 engines with alerts, 2 (4.3%) fired at cycle 1 and are likely false positives from inter-engine baseline variation; the remaining 45 (95.7%) fired at cycle 29 or later. These cycle-1 fires reflect natural inter-engine sensor variation not captured by global noise floors, and are a direction for future work (engine-specific baseline calibration).

The FD001 ablation demonstrates that the improvement of the full pipeline over IF is driven by the z-score component's recall, tempered by the causal component's precision — not by causal conditioning in isolation. To demonstrate causal conditioning as the mechanism, FD002 is required.

4.3 FD002 Regime-Aware Evaluation

The core failure mode being demonstrated is regime mismatch: a reading that is physically normal at cluster 0 (altitude = 42, Mach = 0.84) deviates substantially from means computed at cluster 3 (altitude ≈ 0 , Mach ≈ 0), so a detector trained on single-condition data fires on every engine in a multi-condition fleet.

Table 2b: FD002 Regime-Conditioning Results (259 test engines, alert threshold = 0.3)

Variant	Coverage	Mean (SD)	Median	Wilcoxon- p	Fisher- p
Global z-score (FD001 means)	100%	211.2 (47.8)	203.0	—	—
Regime-aware causal	32%	97.4 (96.4)	35.0	<0.001	<0.001

Lead time in cycles. 259 engines across 6 operating conditions.

The global z-score baseline fires an alert on every single engine in the FD002 test set — 100% coverage that represents a 100% false positive rate. The root cause is regime mismatch: the global means were computed from FD001 training data, which has a single operating condition (op_setting_1 ≈ 0 , op_setting_2 ≈ 0 , op_setting_3 ≈ 100). FD002 includes engines operating at high altitude (op_setting_1 = 42), where HPC outlet temperature and pressure are substantially different from the single-condition training baseline. The global scorer has no way to distinguish this regime-normal variation from engine degradation.

Table 3: FD002 Operating Regime Centroids (KMeans, $k = 6$, trained on op_setting_1/2/3)

Cluster	N (train)	Altitude	Mach	TRA
0	13,458	42.0	0.84	100.0
1	8,122	20.0	0.70	100.0
2	8,002	25.0	0.62	60.0

Cluster	N (train)	Altitude	Mach	TRA
3	8,044	0.0	0.00	100.0
4	8,096	10.0	0.25	100.0
5	8,037	35.0	0.84	100.0

The regime-aware causal system assigns each incoming reading to one of these six clusters and evaluates it against per-cluster regression coefficients. The result is 32% coverage — 68 fewer false alarm engines than the global baseline — with both coverage difference (Fisher $p < 0.001$) and lead-time distribution (Wilcoxon $p < 0.001$) statistically significant. The Wilcoxon result reflects that the regime-aware causal system alerts later and on fewer engines than the global baseline (97.4 vs. 211.2 mean cycles among alerted engines) — lower mean lead time is the intended outcome of false positive suppression, not a performance degradation.

The primary contribution of regime-aware causal scoring on FD002 is **false positive suppression**, not recall improvement. The 32% coverage (82 of 259 engines alerted) reflects the system correctly declining to alarm on regime-normal readings that would otherwise constitute false positives. The lower mean lead time on FD002 (97.4 cycles) compared to FD001 (131.9) is expected: FD002 engines operate across a wider range of conditions, and the causal models conditioned on each regime have less training data per cluster, resulting in somewhat wider residual distributions and later alarm thresholds.

5. Discussion

5.1 What the ablation tells us about mechanism

The full pipeline's Wilcoxon $p = 0.162$ on FD001 is non-significant on lead-time distribution while Fisher $p < 0.001$ on coverage. This is the correct result for homogeneous data: regime conditioning has almost nothing to condition on when op_settings are nearly constant, so the benefit is recall width (2.8× more engines detected) rather than earlier detection timing. The null result on FD001 makes the FD002 finding more credible — a mechanism that only activates where expected is more trustworthy than one that claims improvement everywhere.

5.2 The physics veto as an orthogonal contribution

The structural veto based on the isentropic compression relation is architecturally independent of the causal scoring mechanism. It operates on a different signal (correlation structure between sensor_11 and sensor_15 over a rolling 100-reading window) using a different test (G-test for independence). Its contribution is the detection of sensor faults that would otherwise produce high causal scores — a temperature spike without corresponding pressure increase indicates a sensor malfunction, not engine degradation.

The correct threshold for a 5×5 contingency table is $\chi^2(df = 16, p = 0.05) = 26.30$; the common error of using the $df = 4$ threshold (9.49, applicable to a 3×3 table) would make the veto far more conservative, firing only on extreme decouplings and missing moderate sensor faults that the correct threshold catches.

5.3 Limitations

Engine-specific noise floors. Two engines in the FD001 test set fire alerts at cycle 1, indicating that their healthy initial readings fall outside the global noise floor. Engine-specific baseline calibration (using the first N readings of each engine to set per-engine means) would eliminate these false alarms. This is straightforward to implement but requires a warm-up period before the system is operational.

In-memory state. The forward-fill service and G-test buffer use in-memory data structures. A server restart (which occurs after 15 minutes of inactivity on the free deployment tier) clears per-engine state. For production deployment, these should be backed by Redis.

G-test cold start. The physics veto is inactive for the first 100 cycles of each engine's monitoring history, as the contingency table requires 100 readings to be statistically reliable. For FD001 engines with total run lengths near 130 cycles, this means the veto may fire only in the final 30 cycles before the end of the test record — or not at all for short-life engines. The veto's contribution should be understood as applicable to sustained monitoring scenarios, not to the earliest cycles after system startup.

FD003 and FD004 not evaluated. FD003 has a second fault mode (fan degradation in addition to HPC degradation) under a single operating condition. FD004 combines six conditions and two fault modes. The regime-aware causal architecture is directly applicable but untested on these sub-datasets.

5.4 Future work

- Engine-specific baseline calibration to eliminate inter-engine variation false positives
- Online coefficient update from the human feedback loop (currently the causal coefficients are fixed at deployment)
- Extension to FD003/FD004 fault modes
- Replacement of in-memory state with Redis for production resilience

6. Conclusion

We presented a regime-aware causal anomaly triage system for industrial IoT sensor data and evaluated it on two CMAPSS sub-datasets that isolate complementary failure modes of conventional detectors. On single-condition data (FD001), blended causal and z-score scoring achieves 2.8× better detection coverage than Isolation Forest (47% vs. 17%) at 23% earlier mean lead time (131.9 vs. 107.4 cycles), with a statistically significant coverage advantage (Fisher $p < 0.001$). On multi-condition data (FD002), a global z-score baseline produces a 100% false positive rate from operating-regime mismatch; regime-aware causal scoring suppresses this to 32% coverage with both coverage and lead-time differences statistically significant at $p < 0.001$.

The two experiments provide evidence consistent with regime-aware causal conditioning as the primary driver of false positive prevention on heterogeneous data — not a generically better detector, but a detector that correctly declines to alarm when a reading is normal for its operating context. The physics-based structural veto adds an additional layer of fault discrimination grounded in the isentropic compression relation, detecting sensor malfunctions that statistical scoring alone cannot distinguish from engine degradation.

The system is implemented as an open-source FastAPI backend with a PostgreSQL telemetry store, deployed on Render. All code, data download scripts, and evaluation scripts are available at: <https://github.com/jan-code26/iot-anomaly-triage>

For practitioners, these results suggest a concrete deployment rule: z-score scoring alone suffices when a fleet operates under a single known condition; regime-aware causal scoring is necessary when the fleet spans multiple operating conditions and the cost of false alarms exceeds the cost of reduced recall.

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